Lecture 1 — Foundations of Sustainability, Environmental Challenges and Green Infrastructure

Lecturer notes, slide by slide — with student questions and model answers

Revised version, September 2026 — matches the deck after the lecturer's review notes (learning objectives shortened; Part 3 reframed for business)

Deck: ASM_Lecture 1_Foundations Environmental Challenges and GI_revised.pptx (47 slides)

Built from: original Lectures 1 + 2

Seminar link: Seminar 1 (Environmental challenges of today). At the end of this lecture students choose their Prague case-study site for the whole semester.

What changed in this revision: slide 2 now has seven learning objectives instead of ten, and Part 3 (slides 13–17) presents Green Infrastructure as something a company uses — green roofs, living walls, rain gardens, trees on site — rather than as urban planning. The city-scale view of Green Infrastructure comes later, in Part 6.

How to use this document: read the one-page summary first, then keep the slide-by-slide notes next to you while you teach. Every slide has a short "Say" note in plain words, sometimes an "Ask" question for the room, and a time guide. The last section has questions you can give the students, each with a short model answer.

1. The lecture in one page

The big idea

Sustainability is not charity. It is a way of running a business so that it can keep running. Nature does real work for companies and cities — cooling, cleaning water, stopping floods — and this work has a money value. When a company or a city plans nature on purpose to do that work, we call it Green Infrastructure. In the first half you explain what sustainability is, why companies care, and how a company can use Green Infrastructure on its own buildings and sites. In the second half you show the big environmental problems of our time, how cities make them worse, and how Green Infrastructure helps at city scale. Prague is the example you come back to all semester.

What students should be able to do afterwards

- Explain sustainability in their own words and name the three pillars (people, planet, profit).
- Give three reasons why a company cares about sustainability today.
- Explain what Green Infrastructure is and give four examples a company can use on its own premises.
- Name the main environmental problems of the 21st century and say how cities add to them.
- Explain the urban heat island and what a park does against it.
- Compare grey and green flood protection.
- Pick a Prague green site and describe its environmental, social and economic value.

Words to explain early

Term	Simple meaning
Sustainability	Meeting our needs today without taking away the chance of people in the future to meet theirs.
Triple Bottom Line	Judging a business by three results, not one: people, planet and profit.
Green Infrastructure (GI)	Green roofs, living walls, trees, rain gardens, wetlands and rivers that are planned on purpose to do useful work for a company or a city.
Ecosystem services	The useful things nature does for free: clean water, cooling, food, flood protection, places to relax.
Natural capital	Nature seen as an asset, like a factory or a bank account, that produces these services.
Planetary boundaries	Nine limits of the Earth system. If we cross them, the planet becomes less stable.
Urban heat island	A city is hotter than the countryside around it because concrete and asphalt store heat and there is little shade.
Nature-based solution	Solving a city problem (heat, floods) with nature instead of only pipes and concrete.

Timing plan

The plan below assumes a long block (about 3.5 hours including a break). If your slot is shorter, skip the slides marked optional first.

Block	Slides	Minutes
Welcome, objectives, agenda	1–3	10
Part 1 — Why sustainability matters (incl. opening discussion)	4–8	25

Part 2 — Sustainability and business	9–12	20
Part 3 — Green Infrastructure in business	13–17	20
Prague focus and Singapore case, mid-lecture discussion	18–23	20
Break	24	15–20
Part 4 — Planetary challenges	25–30	25
Part 5 — Cities and nature	31–34	15
Part 6 — Green Infrastructure as a solution	35–41	25
Classroom workshop and seminar preparation	42–43	25
Key messages, reading, questions	44–46	10

Before class

- Check the lecturer name on slide 1.
- Decide the list of Prague sites students may choose from (slide 43) and whether a group can pick a site not on the list.
- Have the seminar dates ready — students will ask when Seminar 1 is.
- Optional: bring one recent news item about a heat wave, flood or drought in Czechia for slide 27 or 28.
- Optional: think of one company building in Prague with a green roof or a living wall that students may know — it makes Part 3 concrete.

2. Slide-by-slide notes

Numbers match the slide numbers in the PowerPoint file. **Say** = what to tell the students, in your own words. **Ask** = a question to put to the room. **Tip** = practical advice.

1 Title slide — 2 min

Say: Welcome the students. Say who you are and that this course is about sustainability as a management skill, not only as an environmental topic. Tell them the course has six lectures and six seminars, and a semester project about a real place in Prague.

2 Learning Objectives — 3 min

Say: Read the seven points slowly. Tell them the first half is about ideas, business and what a company can do with nature; the second half is about problems and solutions in cities. They do not have to remember everything today; the same ideas will come back in every lecture.

> **Tip:** The last point ("urbanization, natural systems and Green Infrastructure") is the bridge between the two halves — say that the second half is where it gets answered.

3 Today's Lecture Agenda — 2 min

Say: Show the structure: six parts with a break in the middle. Point to the workshop near the end — say they will choose a Prague site today that they will work on all semester.

> **Tip:** Say when the break will be. Students concentrate better when they know.

Part 1 — Why Sustainability Matters (slides 4–8)

4 Part 1 — Why Sustainability Matters — 1 min

Say: Just announce the part. No content here.

5 Opening Discussion — 10 min

Say: Do not lecture yet. Ask the three questions on the slide and let students talk. Write their words on the board. You will use their answers on the next slide.

> **Ask:** What does sustainability mean to you? Can a company be profitable and sustainable at the same time?

> **Tip:** If the room is quiet, ask them to talk to their neighbour for two minutes first, then collect answers. Accept every answer — the goal is to hear what they already think.

6 What is Sustainability? — 5 min

Say: Read the Brundtland definition from 1987. It is the most used definition in the world. Then explain the three ideas behind it: think long term, use resources carefully, and think about people who are not born yet. Connect it to what students said on the previous slide.

7 The Triple Bottom Line

5 min

Say: A normal company counts one result: profit. The Triple Bottom Line says count three: people, planet and profit. Give an example for each column. For example, fair wages is a people result, lower carbon is a planet result, innovation is a profit result.

> **Ask:** Can you think of a company that is strong in one column and weak in another?

8 Evolution of Corporate Sustainability

4 min

Say: Tell the story in three steps. In the 1970s companies only followed the law. In the 1990s they started voluntary programmes and reports. Today, for companies like Unilever or IKEA, sustainability is part of the main strategy. The point: the topic moved from the legal department to the board room.

Part 2 — Sustainability and Business (slides 9–12)

9 Part 2 — Sustainability and Business

1 min

Say: Announce the part: now we look at why companies actually do this.

10 Why Companies Care About Sustainability

6 min

Say: Four groups push companies: regulators, investors, customers and employees. Go through them one by one. Regulation: the EU Green Deal and the reporting rules (CSRD). Investors: big funds ask about climate risk before they invest. Customers: many people choose greener products. Employees: young people want to work for companies they respect.

> **Ask:** Which of the four do you think pushes hardest in Czechia today?

11 Business Risks of Unsustainability

4 min

Say: Turn it around: what happens if a company ignores this? Floods and heat can stop production. Scarce resources become expensive. And if a company claims to be green but is not, the media and customers punish it — this is called greenwashing. We will return to greenwashing in Lecture 5.

12 Case Study: Unilever

5 min

Say: Unilever is a large consumer-goods company. It works on where its raw materials come from, on using less water, and on lower carbon along the whole chain. Ask the discussion question and let two or three students answer.

> **Ask:** How does sustainability create business value for a company like this?

Part 3 — Green Infrastructure in Business (slides 13–17)

13 Part 3 — Green Infrastructure in Business

1 min

Say: Announce the part. Say this is the key idea of the whole course, so they should pay attention now. Add one sentence of framing: in this part we look at Green Infrastructure from the point of view of a company — what it can build on its own roof, walls and land, and why that pays. The city-wide view comes after the break.

14 What is Green Infrastructure?

5 min

Say: Read the definition. Then explain it in plain words: we usually think of infrastructure as roads, pipes and cables — things a business invests in because they do a job. Green Infrastructure means treating nature the same way. For a company that means a green roof, a living wall, a rain garden in the car park or trees around the site: nature used as a business asset that cuts operating costs, manages risk and delivers services the company would otherwise pay for. Point to the living wall photo: nature built into an office building.

> **Ask:** What does a company normally pay for that a tree or a green roof could do for free? Guide them to: cooling, shade, rainwater handling, a nicer place to work.

> **Tip:** Write 'GI' on the board — you will use this short form all semester.

15 Green Infrastructure a Company Can Use

4 min

Say: Four things a company can put on its own premises, each with a business benefit. Green roofs: lower cooling bills, the roof lasts longer, lower stormwater fees. Living walls: cooler buildings, cleaner air, a strong brand image. Rain gardens: flood protection for the site, lower drainage costs, easier compliance with water rules. Trees on site: shade and cooling, staff well-being, carbon absorption. The key word is benefit — each one does work the company would otherwise pay for.

> **Tip:** If you know a Prague office or shopping centre with a green roof or living wall, name it here. A local example makes the list real.

16 Green Roofs as a Business Investment

3 min

Say: The photo is a green roof in Chicago, in the centre of a dense city. Use it to show the business case, not the city planning: for the building owner it lowers cooling costs, extends the life of the roof because the plants protect the membrane from sun and frost, and reduces stormwater charges. A company makes exactly the same case for its own offices, warehouses and factories. Say that it costs more to build but pays back over the life of the roof — we will do the full-life calculation in Lecture 4.

17 Natural Capital and Ecosystem Services

6 min

Say: Nature gives us four kinds of services. Provisioning: things we take, like food, water and raw materials. Regulating: processes nature controls, like flood control, climate and pollination. Cultural: recreation and tourism. Supporting: the basic processes underneath, like soil formation and nutrient cycling. All of these have a money value, even if nobody sends a bill. Then read the box at the bottom and make it about a company: provisioning services are the raw materials and water in its supply chain; regulating services protect its sites from floods and heat; cultural and supporting services are behind the reputation and the licence to operate a company gets when it protects nature.

> **Ask:** Pick a company you know. Which of the four services does it depend on most? Guide them: a brewery depends on water (provisioning), a logistics firm on flood protection (regulating), a hotel on landscape and recreation (cultural).

Prague Focus and case study (slides 18–23)

18 Prague Focus

1 min

Say: Announce that we now look at our own city.

19 Prague — a City Built Around its River

2 min

Say: Show the photo. The Vltava is the blue-green spine of Prague. Everything in this course can be seen along this river: floods, tourism, property values, wildlife.

20 Prague as a Living Laboratory

5 min

Say: Name the four natural assets: Stromovka, Divoká Šárka, Prokopské údolí and the river corridor. Ask students which they know. Then ask the discussion question: how do these places support the city environmentally, socially and economically? Collect one answer for each of the three.

> **Tip:** Most students know Stromovka and Letná. Fewer know Prokopské údolí — a short description helps.

21 Human–Nature Interaction

4 min

Say: Three questions about people and green space. Focus on the property-value question: flats next to a park are more expensive. This is the simplest proof that nature has a money value.

> **Ask:** Why are flats next to a park more expensive?

22 Case Study: Singapore — City in Nature

optional 4 min

Say: Singapore is one of the densest cities in the world and still built nature into it: green roofs, vertical gardens, and the Supertrees in the photo. The lesson: density is not an excuse. If Singapore can do it, Prague can.

23 Mid-Lecture Discussion — 6 min

Say: Three questions to close the first half. Take the first one seriously: is sustainability a cost or an investment? Let students argue both sides for a few minutes. You do not need to give a final answer — the course will.

> **Tip:** If you are short of time, ask only question 1.

24 Break — 15–20 min

Say: Tell students the second half starts with the big global problems and then comes back to Prague. Say the exact time you restart.

Part 4 — Planetary Challenges (slides 25–30)

25 Part 4 — Planetary Challenges — 1 min

Say: Announce the part. Say: in the first half we saw why sustainability matters to business; now we see the problems themselves.

26 The Planetary Boundaries Framework — 6 min

Say: Scientists at the Stockholm Resilience Centre defined nine limits for the Earth system. Inside the limits the planet stays stable; outside, it may not. The slide shows the ones under most pressure: climate, biodiversity, fresh water, land use and pollution by new chemicals. Several of these limits are already crossed.

> **Ask:** What happens when a boundary is crossed? Guide them to the answer: the system can change suddenly and does not simply return when we stop.

27 Climate Change — 4 min

Say: Keep it simple: burning fossil fuels, cutting forests and industry put greenhouse gases into the air; the planet warms; we get more heat waves, floods and droughts. The next two slides show what this looks like.

28 Drought: a Growing European Risk — 2 min

Say: Photo of a reservoir at a historic low. Say that drought is no longer only a southern problem — Central Europe, including Czechia, has had serious dry years. Farmers, breweries and power plants all need water.

29 The Biodiversity Crisis — 5 min

Say: Biodiversity means the variety of living things. It is falling fast because of habitat loss, cities spreading, pollution and invasive species. Why should a manager care? Because food, water, pollination and a stable climate all depend on it. Point out that urban expansion is on the list — cities are part of the problem.

30 Land-Use Change Seen from Above

optional 2 min

Say: Satellite photo of deforestation. Use it to show that land-use change is visible from space — and that this is one of the planetary boundaries from slide 26.

Part 5 — Urban Development and Nature (slides 31–34)

31 Part 5 — Urban Development and Nature

1 min

Say: Announce: now we zoom in from the planet to the city.

32 Urbanization Trends

4 min

Say: More than half of all people now live in cities, and almost all future growth will be in cities. This creates demand for housing and transport and eats green space. So cities are where sustainability will be won or lost.

33 Urban Heat Islands

5 min

Say: Explain the heat island in three steps: asphalt and concrete store heat; there is little shade and little water to evaporate; buildings and cars add waste heat. The result: the city centre can be several degrees hotter than the edge on a summer night. This is a health problem for old people and a cost problem for anyone who pays for cooling.

34 Prague Example: Green vs Dense Urban Areas

4 min

Say: Compare Stromovka with the city centre. Under the trees it is noticeably cooler on a summer day. Ask students where in Prague they feel this difference themselves.

> **Ask:** Where in Prague do you feel the heat island most? Where do you go to cool down?

Part 6 — Green Infrastructure as a Solution (slides 35–41)

35 Part 6 — Green Infrastructure as a Solution

1 min

Say: Announce: the problems are clear; now the solutions. Say that in Part 3 we saw Green Infrastructure on one company's site; now we see the same idea at the scale of a whole city.

36 Nature-Based Solutions

3 min

Say: A nature-based solution means solving a problem by working with nature instead of against it. Urban forests, green roofs and wetlands are examples. Connect back to slide 15 — the same kinds of things a company can build, now used by a city as tools against heat and floods.

37 Flood Management: Grey vs Green

5 min

Say: Grey means pipes and concrete channels: they do one job and cost a lot to maintain. Green means wetlands, rain gardens and landscapes that hold water: they do the flood job and also cool the area, give recreation and support wildlife. One investment, several benefits. This comparison is a main argument of the whole course.

38 Smart City Example: Copenhagen

4 min

Say: Copenhagen had very heavy rainstorms ('cloudbursts'). Instead of only building bigger sewers, the city designed parks and streets that flood safely and hold the water. Result: flood protection and better public space at the same time. The photo shows the harbour bath — the harbour water is now clean enough to swim in.

39 Smart City Example: Rotterdam

4 min

Say: Rotterdam's water squares are playgrounds and sports courts in dry weather and water basins in heavy rain. Ask the question on the slide: can infrastructure serve more than one purpose? The answer is yes, and that is the point of GI.

> **Ask:** What else in a city could do two jobs at once?

40 When the River Rises

3 min

Say: Photo of the Vltava flood in 2013. Remind students of 2002 and 2013: floods close shops, metro lines and offices for weeks. Flood resilience is a business issue in Prague, not only a city-hall issue.

41 Prague Case Study: Vltava River Corridor

5 min

Say: Ask what the river corridor gives the city, using the three columns: environmental (flood space, wildlife corridor), social (tourism, recreation), economic (property value, attractiveness). This is the same three-value structure students will use in the workshop and in their project.

Workshop, seminar preparation and closing (slides 42–46)

42 Classroom Workshop

20 min

Say: Groups of three or four. Each group takes one Prague location and lists its environmental, social and economic benefits and its sustainability challenges. Give them 12 minutes to work and 8 minutes to share. Details are in section 3 of these notes.

43 Preparing for Seminar 1

5 min

Say: Explain that each group must choose one Prague site from the list and will work with it all semester: analysis, SWOT, stakeholders, indicators and finally a proposal. Ask them to decide before Seminar 1. Tell them the date of Seminar 1.

> **Tip:** Try to avoid five groups choosing Stromovka. If needed, allow a maximum of two groups per site.

44 Key Messages

4 min

Say: Read the eight messages. They are the summary of the whole lecture. Ask if anything is unclear.

45 Recommended Reading

2 min

Say: Point to the compulsory reading (the Business in the Community handbook). The other books are for students who want more. Say what the next lecture is about: pollution, climate resilience and the circular economy.

46 Any questions?

5 min

Say: Take questions. If there are none, ask one of the student questions from section 5 of these notes.

47 Image credits

Say: You do not need to show this slide. It is there for copyright reasons — keep it in the file.

3. How to run the class activities

Opening discussion (brainstorm) (slide 5, 10 minutes)

1. Show the three questions. Give students two minutes to talk in pairs.
2. Collect answers from four or five pairs. Write key words on the board.
3. Do not correct anyone. Say you will come back to their words during the lecture.
4. On slide 6, connect the Brundtland definition to words they used (for example 'future', 'balance', 'resources').

Classroom Workshop — analyse one Prague location (slide 42, 20 minutes)

1. Make groups of three or four. Each group picks a Prague location (Stromovka, Letná, Vltava riverbank, Divoká Šárka, Prokopské údolí, Hostivař, Troja or one they suggest).
2. 12 minutes: the group writes four lists — environmental benefits, social benefits, economic benefits, sustainability challenges. Two or three points per list is enough.
3. 8 minutes: each group reads out one benefit and one challenge. Write the challenges on the board — they are the seeds of the semester project.
4. Close by saying: the site you chose today can be your project site, or you can change your mind before Seminar 1.

> **Watch for:** Groups often forget the economic column. Prompt them: who earns money because this place exists? Cafés, tourism, flats, events.

4. If a student asks…

“Isn't sustainability just marketing?”
Sometimes yes — that is greenwashing, and we will study it in Lecture 5. But regulation, investors and physical risks like floods are real. A company that ignores them loses money. The test is always: can you measure it?

“If the Earth's limits are already crossed, what is the point?”
Crossing a boundary makes things worse, but every step back reduces the damage. Cities are where the biggest gains are possible, because that is where most people live. That is why this course focuses on cities.

“Why should a business pay for a park that everyone uses?”
Good question — we debate exactly this in Lecture 3. Short answer for now: companies benefit from cooler streets, fewer floods, happier employees and higher property values, so they often gain more than they pay.

“Is a green roof not very expensive?”
It costs more to build, but it lasts longer, saves energy and holds rainwater. In Lecture 4 we compare the whole life cost of a green roof and a normal roof.

“Our company rents its office. What can we do with Green Infrastructure?”
Smaller things still count: plants and a living wall inside, trees and a rain garden in the car park, or asking the landlord for a green roof at the next renovation. Many companies also support Green Infrastructure off-site — for example paying for trees or wetlands near their supply chain. Lecture 3 looks at these business models.

5. Questions for the students — with model answers

Use them at the end of the lecture, as homework, or as a bank for the written test. Tags: recall checks memory, understand checks the idea, apply asks students to use it on a real case. The answers are short — a good student answer can be longer.

1. Explain the Brundtland definition of sustainable development in your own words. recall

Model answer: Development that meets the needs of people today without taking away the ability of future generations to meet their own needs. Key ideas: long-term thinking, careful use of resources, responsibility for the future.

2. What are the three parts of the Triple Bottom Line? Give one example of each for a food company. understand

Model answer: People (social), planet (environmental), profit (economic). Examples: fair pay for farmers; lower water use in production; profit from selling a trusted brand.

3. Name four reasons why a company today cares about sustainability. recall

Model answer: Regulation (EU Green Deal, reporting rules), investors (they check climate and ESG risk), customers (demand for greener products), employees (want to work for responsible companies).

4. What is Green Infrastructure? Give four examples a company can use on its own premises and say what each one does for the company. recall

Model answer: A planned network of natural and semi-natural areas that gives services; for a company, nature used as a business asset. Examples: green roof (lower cooling bills, longer roof life, lower stormwater fees), living wall (cooler building, cleaner air, brand image), rain garden (flood protection, lower drainage costs), trees on site (shade, staff well-being, carbon absorption).

5. Explain the four types of ecosystem services and give one example of each from Prague. understand

Model answer: Provisioning (things we take: water from the Vltava), regulating (processes nature controls: the river corridor holding flood water, trees cooling streets, pollination), cultural (recreation and tourism: Letná, Stromovka), supporting (basic processes: soil formation and nutrient cycling in parks and forests).

6. What is an urban heat island and what are its three main causes? recall

Model answer: The city is hotter than the surrounding countryside. Causes: asphalt and concrete store heat; little vegetation and shade; waste heat from buildings and traffic.

7. Why is biodiversity loss a problem for business, not only for nature? understand

Model answer: Food production, clean water, pollination and a stable climate all depend on healthy ecosystems. If they fail, raw materials become scarce and expensive, and some business models stop working.

8. Compare grey and green flood protection. Give two advantages of each. understand

Model answer: Grey (pipes, concrete channels): predictable, works immediately, well understood by engineers. Green (wetlands, rain gardens, water squares): several benefits from one investment (cooling, recreation, wildlife), often cheaper to maintain, gets better as plants grow. Best answers say a city usually needs both.

9. Describe what Rotterdam's water squares do in dry weather and in heavy rain. Why is this a good example of Green Infrastructure thinking? apply

Model answer: Dry weather: playground and sports court. Heavy rain: a basin that holds water so the sewers do not overflow. It shows that one piece of infrastructure can do two jobs, which is the core idea of GI.

10. Choose one Prague green site. Describe its environmental, social and economic value and one sustainability challenge it faces. apply

Model answer: Any well-argued answer. Example — Stromovka: environmental (cooling, habitat, rain absorption), social (recreation, events, health), economic (higher flat prices nearby, cafés, tourism), challenge (pressure from events and visitors, drought stress on old trees, traffic around it).

11. A company owns a warehouse with a large flat roof and a car park. Suggest two Green Infrastructure investments and give one business benefit of each. apply

Model answer: Any sensible pair. Example: a green roof (lower cooling costs in summer, longer roof life, lower stormwater fees) and a rain garden or permeable surface in the car park (less flooding on site, lower drainage costs, easier compliance). Trees around the site (shade for parked vehicles, better working environment) is also a good answer.

Handout version (questions only)

1. Explain the Brundtland definition of sustainable development in your own words.
2. What are the three parts of the Triple Bottom Line? Give one example of each for a food company.
3. Name four reasons why a company today cares about sustainability.
4. What is Green Infrastructure? Give four examples a company can use on its own premises and say what each one does for the company.
5. Explain the four types of ecosystem services and give one example of each from Prague.
6. What is an urban heat island and what are its three main causes?
7. Why is biodiversity loss a problem for business, not only for nature?
8. Compare grey and green flood protection. Give two advantages of each.
9. Describe what Rotterdam's water squares do in dry weather and in heavy rain. Why is this a good example of Green Infrastructure thinking?
10. Choose one Prague green site. Describe its environmental, social and economic value and one sustainability challenge it faces.
11. A company owns a warehouse with a large flat roof and a car park. Suggest two Green Infrastructure investments and give one business benefit of each.

Lecturer notes for Lecture 1 — Foundations of Sustainability, Environmental Challenges and Green Infrastructure (revised after the lecturer's review notes).
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